

STEPCUE · REQUIREMENTS

StepCue requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/stepcue		
Source	docs/03-requirements.md @ 84be87d		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

StepCue requirements

These are first-pass requirements for the concept. Targets are proposals for review, awaiting Amish. The status column gives the TRL 2 position from the estimates in the design precis (STC-PRC-001); nothing here has been measured.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Sense plantar loading at the main contact regions	5 or more pressure sites (heel, lateral midfoot, first and fifth metatarsal heads, hallux), each sampled at 100 Hz or more	Design review; firmware configuration	Met by design
R2	Sense shank or foot motion for the freeze band	6-axis IMU at 100 Hz or more, covering the 3 to 8 Hz freeze band and the 0.5 to 3 Hz locomotor band	Datasheet; sampling calculation	Met by design (heel pod IMU)
R3	Detect freezes quickly	Median delay from freeze onset to detection 2 s or less	Offline replay of labeled data	Not demonstrated. Needs labeled data
R4	Detect freezes reliably	Episode sensitivity 80 % or more; window specificity 85 % or more	Offline replay of labeled data	At risk. Published pressure-only results are about 77 to 80 % sensitivity and 83 to 85 % specificity
R5	Avoid nuisance cues	1 false cue or fewer per 10 min of ordinary walking	Offline replay; later supervised trial	Not demonstrated
R6	Start the cue promptly	Cue starts 0.1 s or less after a detection	Firmware timing calculation	Met by estimate (about 20 ms)
R7	Deliver a rhythmic cue the wearer can feel or hear	Haptic pulses at the wearer's baseline cadence (adjustable 60 to 130 per minute); audio beeps of 60 dB(A) or more at the ear	Calculation from datasheets; later bench measurement	Haptic met by estimate; audio at risk (about 60 dB(A) at 1.5 m, estimate)
R8	Stop cueing when walking resumes	Cue stops within 3 regular steps, or after 15 s at most	Firmware logic review	Met by design
R9	Keep the shoe comfortable	Insole stack 5.0 mm or less; no rigid part thicker than 1 mm under the heel or metatarsal heads	Massing model; later pressure mapping	Met, no margin (5.0 mm)
R10	Keep the heel pod light and small	Pod 35 g or less; no larger than 45 x 40 x 20 mm	Massing model; later weighing	Met by estimate (about 28 g; 44 x 38 x 19 mm)
R11	Run a full day and more	2 days or more between charges at 16 h per day of wear	Power budget calculation	Met by estimate (about 3 days)
R12	Usable with reduced dexterity	Pod clips on and off one-handed; USB-C charging with the pod off the shoe; one large button to pause cues	Design review; later user session	Met by design, not user-tested
R13	Keep data with the wearer	Detection and cueing on the device; optional event log exported over USB or BLE by the owner only; no cloud	Design review	Met by design
R14	Low cost and buildable	Parts cost \$200 or less per unit, with one instrumented insole; no custom rigid PCB for the first build	Priced BOM	Met by estimate (about \$106); a pair (about \$212) would not meet it

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R15	Safe to wear	Protected lithium cell outside the shoe; no exposed conductors; skin-contact materials suitable for footwear	Design review	Met by design

■ Requirements not met or not yet demonstrated

- **R3, R4 and R5** (detection delay, reliability and nuisance cues) cannot be shown on paper. They need a labeled plantar-pressure and IMU data set, and published results suggest R4 is at the edge of what pressure data alone achieves.
- **R7 audio** is at risk: a small piezo at the heel gives about 60 dB(A) at the ear by estimate, which may be masked outdoors. A phone or earbud route would meet it.
- **R9** has no thickness margin.
- **R14** is met only for one insole. A pair would exceed the \$200 budget.

■ Assumptions

- Freeze-related leg trembling sits in a 3 to 8 Hz band, compared with 0.5 to 3 Hz for normal stepping; this is the basis of the widely used freeze index ([Sweeney et al., 2019](#); [freeze index reliability, JNER 2023](#)).
- Cueing near the person's own cadence helps, while cueing well below it can make freezing worse ([Sweeney et al., 2019](#), summarizing Arias and Cudeiro 2010 and Lee et al. 2012).
- Shoe size about EU 42 (US men's 8.5) for the massing model; the insole is cut to size in practice.
- Wear of 16 h per day with overnight charging.